

ABIRAMI B

AI Developer

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CAREER OBJECTIVE

Passionate AI developer with hands-on experience in building intelligent chatbots, leveraging Natural Language Processing (NLP) and Large Language Models (LLMs) to create innovative solutions. Eager to contribute to cutting-edge AI projects, and thrive in environments that foster continuous learning and growth.

EXPERIENCE

Yectra Technologies | AI Developer Ongoing

Coimbatore | 2024 - Present

Chatbot development for web application:

Leveraged a pre-trained Large Language Model (LLM) from the Hugging Face Transformers library and fine-tuned it for a domain-specific conversational AI chatbot using advanced fine-tuning techniques. Applied quantization methods, including Parameter-Efficient Fine-Tuning (PEFT), Low-Rank Adaptation (LoRA), and Quantized Low-Rank Adaptation (QLoRA), to enhance model performance. Integrated the fine-tuned model with a REST API, enabling seamless user interactions and optimizing question-answering capabilities.

RAG-Based Chatbot with Memory:

Developed a history-aware Retrieval-Augmented Generation (RAG) chatbot by integrating external data. Utilized embedding techniques and stored vectorized data in Chroma and Pinecone databases. Applied chunking strategies tailored to specific use cases. Implemented advanced memory management to retain key context and enhance response accuracy with better prompting strategies and personalization.

Software Utility Marketplace:

Designed and developed AI-driven utilities to enhance functionality and usability within a software marketplace.

Implemented a QR Code Generator for efficient encoding and decoding of information, enabling seamless data sharing and integration.

Engineered a Paragraph Summarizer utilizing advanced natural language processing (NLP) algorithms to deliver precise and concise summaries, optimizing content accessibility and user productivity.

PROJECTS

Disaster Classification of Crowd-Sourced Tweets Using Deep Learning

Developed a multimodal deep learning classification model that analyzes both text and image data from crowd-sourced tweets. The model classifies disaster events, contributing to more effective disaster response and management.

Amazon Clone Website

Built a front-end Amazon clone website, demonstrating expertise in web development. The website features a visually appealing and user-friendly interface to provide an immersive shopping experience.

SKILLS

Core Development:

- Programming Languages: Python, Java, HTML, CSS
- Tools / Platforms: Postman, Docker Desktop, VS Code
- Version Control Systems: Git

Data & Cloud:

- Database: NoSQL (Azure Cosmos DB), MongoDB, Chroma
- Cloud Services: Azure
- Data Processing Frameworks: numpy, pandas

AI & Machine Learning:

- Machine Learning Frameworks: TensorFlow, PyTorch, Keras
- Generative AI Models: Transformers
- AI/ML Libraries: Hugging Face Transformers

API & Backend:

- REST API: Flask, FastAPI

Specialized AI Skills:

- Fine Tuning: Adapting LLMs for domain-specific tasks
- RAG: Combining LLMs with external data retrieval systems

EDUCATION		
Dr.NGP Institute of Technology		
B.Tech in Information Technology		2020 – 2024
CGPA: 87.4%		
GRG Matriculation Higher Secondary School		
Higher Secondary Certificate (HSC)		2019 – 2020
Percentage: 70%		
GRG Matriculation Higher Secondary School		
Secondary School Leaving Certificate (SSLC)		2017 – 2018
Percentage: 89%		